

# Sourav Ray

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## Professional Summary

Highly experienced Data & AI Engineer with 3+ years of experience and an M.Tech degree in Operations Research, leveraging production data engineering, pipeline reliability, and AI/ML capabilities to drive business impact. Proven track record of designing and implementing scalable architectures, including containerized Python frameworks, metadata-driven automation, and multi-stage reconciliation pipelines. Skilled in modern AI architectures such as LLMs and deep learning, with expertise in tools like AWS, Snowflake, GCP, and PySpark.

## Highlights

- Promoted within 2 years for consistently exceeding performance expectations, driving high-impact Data Engineering and GenAI/Agentic AI initiatives.
- 3+ years of experience building Snowflake-based Data Mesh, Governance, and GenAI solutions, spanning both core data platform engineering and applied AI/ML delivery.
- Built multi-agent, MCP-based AI orchestration systems integrating LLMs with enterprise platforms (ServiceNow, Snowflake, observability tools) for automated investigation and root-cause analysis.
- Hands-on experience with LangChain, RAG, vector search, and LLM-based inference for schema mapping, semantic search, and automated RCA.
- SnowPro Advanced Architect, SnowPro Advanced Data Scientist, and AWS Solutions Architect Associate certified.
- Proven track record delivering enterprise-scale data quality, governance, and automation platforms; mentored junior engineers on DBT development, data quality frameworks, and GenAI-based automation best practices.

## Technical Skills

**Cloud & Data Platforms:** AWS (S3, Lambda, Kinesis, Firehose, DMS), Snowflake (Snowpark, Snowpipe), GCP (Vertex AI, BigQuery)

**Data Engineering:** DBT, Apache Airflow, Coalesce, SQL, PySpark, Kafka, ETL/ELT, Data Contracts, Data Mesh

**AI / ML & GenAI:** OpenAI, Gemini, LangChain, RAG, ChromaDB, NLP, TensorFlow, Time-Series Forecasting, Genetic Algorithms, Multi-Agent Orchestration, Model Context Protocol (MCP)

**Programming:** Python, SQL, Shell Scripting

**Tools:** Docker, Kubernetes, Git, GitHub Actions, Sigma

## Professional Experience

**Factspan Analytics, Bengaluru**

*Data & AI Engineer*

Sep 2023 – Present

**Customer 360 Data Mesh & Governance | Snowflake, AWS, DBT**

- Architected a containerized Python Data Contract framework supporting RDBMS, flat-file, and streaming ecosystems, improving enterprise data accessibility by 60%.
- Engineered metadata-driven automation generating Silver and Gold DBT artifacts (.sql, .yaml, .md), reducing development effort by 80%.
- Designed ingestion frameworks using AWS DMS, Kinesis, Firehose, S3, and Snowpipe for near real-time onboarding.
- Built multi-stage reconciliation pipelines using row-count validation and MD5-based record verification to ensure data fidelity.
- Developed automated RCA tooling identifying column-level discrepancies and format anomalies.
- Validated SAP BusinessObjects reports against Snowflake models ensuring reporting consistency across legacy and modern platforms.
- Governed finance datasets including GL, revenue, and transactional domains through audit-ready validation controls.

**AI-Powered Network Incident Orchestrator | Python, MCP, LLM Agents**

- Built a multi-agent orchestrator that coordinates specialized AI agents to accelerate network incident investigation and root cause analysis (RCA) across ServiceNow, Infoblox, NetBox, SevOne, Datadog, Splunk, and LogicMonitor.
- Designed per-platform agents, each backed by its own Model Context Protocol (MCP) server, providing a consistent abstraction layer over heterogeneous observability tools and device-management APIs, including secure SSH-based access to Cisco and non-Cisco network devices.
- Implemented dynamic agent selection and parallel invocation logic that correlates multi-source data – incidents, logs, metrics, topology, device health, and configuration – into a single consolidated RCA report.
- Improved performance and scalability through in-memory caching and per-agent concurrency controls (capped at 3 concurrent users), reducing Mean Time to Detect (MTTD) and Mean Time to Resolve (MTTR).

**GenAI Schema Validator & Automated Mapping Platform**

- Developing an enterprise-scale schema and report matching platform using Python, Vector Search, and LLMs, implementing Exact Match, Semantic Match, and LLM-based inference layers for automated column mapping.
- Built governance controls to identify and exclude masked PII attributes, and leveraged LLMs to auto-generate metadata descriptions for undocumented fields.

- Leading UAT and validation across complex enterprise datasets and reporting structures.

#### **Data-Powered Assist Chatbot**

- Developed Gold-layer models supporting Sigma lineage analytics.
- Enabled semantic querying in Snowflake through NLP-powered chatbot workflows.
- Built ETL pipelines using Coalesce and integrated image URI-based workflows.

#### **Education**

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**NIT Durgapur** | M.Tech, Operations Research | CGPA: 8.36/10 | 2021–2023

**GCETTB** | B.Tech, Textile Technology | CGPA: 7.43/10 | 2015–2019

#### **Certifications**

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- Snowflake SnowPro Advanced Architect
- Snowflake SnowPro Advanced Data Scientist
- AWS Certified Machine Learning – Specialty
- AWS Certified Solutions Architect – Associate